Lab Manual - LAB 10

Semester V – Fall

Group A and B

Course Title: Computer Networks

Bachelor of Science in Software Engineering

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LAB 10: Star, Ring, and Mesh Topologies in Cisco Packet Tracer

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Ring, Mesh, and Star Topologies

Learning Objectives

By the end of this lab, students will be able to:

- Explain the structure of star, ring, and mesh topologies.
- Construct each topology using Cisco Packet Tracer.
- Connect PCs and switches using appropriate Ethernet connections.
- Configure IP addresses to test communication between devices.
- Observe how network design affects connectivity and redundancy.
- Compare the advantages and limitations of different topologies.

Introduction to Network Topology

A **network topology** describes how devices are arranged and connected in a computer network. There are two ways to understand a topology:

- **Physical topology:** How devices and cables are physically arranged.
- **Logical topology:** How data moves between devices through the network.

In this lab, we will construct three topologies: star, ring, and mesh.

Required Devices

Device	Quantity	Purpose
PCs	4	End devices that send and receive data
Cisco 2960 switches	4	Connect devices within the network
Ethernet cables	As required	Connect PCs and switches

Use the standard Copper Straight-Through cable to connect PCs to switches. For switch-to-switch connections, Packet Tracer's automatic connection option or a suitable Ethernet cable can be used.

Activity 1: Star Topology

In a **star topology**, every end device connects to a central device, such as a switch. The central switch acts as the main connection point for all the computers.

Example: A computer laboratory where every PC connects to a central switch.

Advantages

- Easy to install and manage.
- Adding or removing a PC is straightforward.
- Failure of one PC's cable usually affects only that PC.
- Faults can be easier to identify.

Limitations

- Failure of the central switch can disconnect all attached devices.
- More cable may be required than in some simpler arrangements.

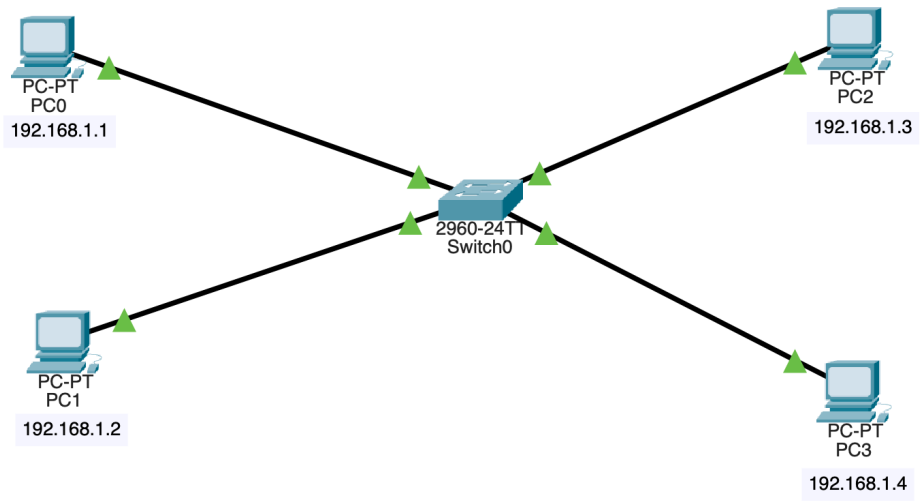
Practical Implementation in Packet Tracer

Step 1: Create the topology

1. Open Cisco Packet Tracer.
2. Create a new project.
3. Add one Cisco 2960 switch.
4. Add four PCs.
5. Arrange the switch in the center and the PCs around it.

Step 2: Connect the devices

Connect each PC's FastEthernet interface to a separate port on Switch0.



Step 3: Configure IP addresses

Open each PC and navigate to **Desktop → IP Configuration**.

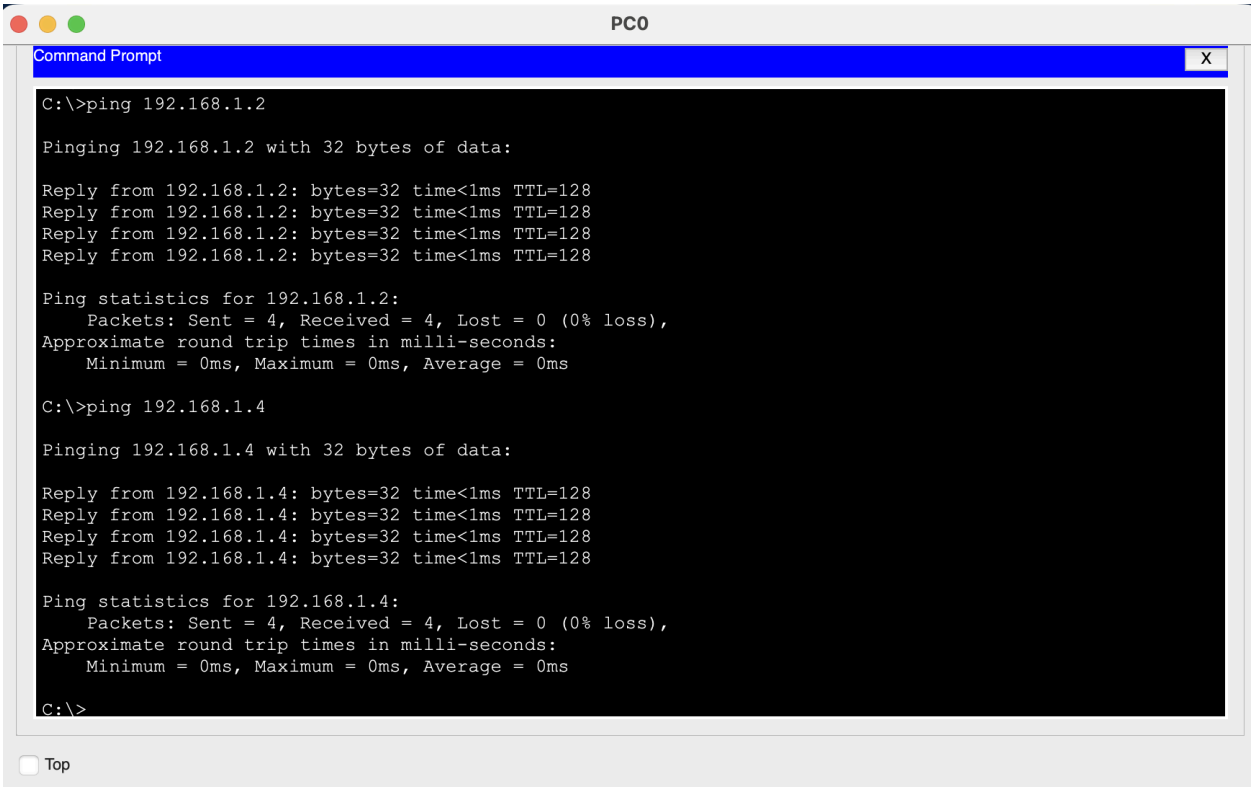
Use the following settings:

PC	IPv4 Address	Subnet Mask
PC0	192.168.1.1	255.255.255.0
PC1	192.168.1.2	255.255.255.0
PC2	192.168.1.3	255.255.255.0
PC3	192.168.1.4	255.255.255.0

Leave the default gateway blank because all four PCs are on the same local network and no router is required for this activity.

Step 4: Test connectivity

1. Open PC0.
2. Select **Desktop → Command Prompt**.
3. Enter: ping 192.168.1.2



Activity 2: Ring Topology

In a **ring topology**, devices are connected in a closed loop. Each device has connections to two neighboring devices in the ring. Data can travel around the ring from one device toward another. The exact forwarding method depends on the technology used.

Example: A ring-based network connecting devices or sites around a campus.

Advantages

- Provides a defined circular connection path.
- Can offer an alternative path in some ring technologies.
- Can be useful where ring protocols provide resilience.

Limitations

- A break in a simple, unprotected ring can interrupt communication.
- Adding or removing devices may require changes to the ring.

- Troubleshooting can be more difficult than in a simple star network.

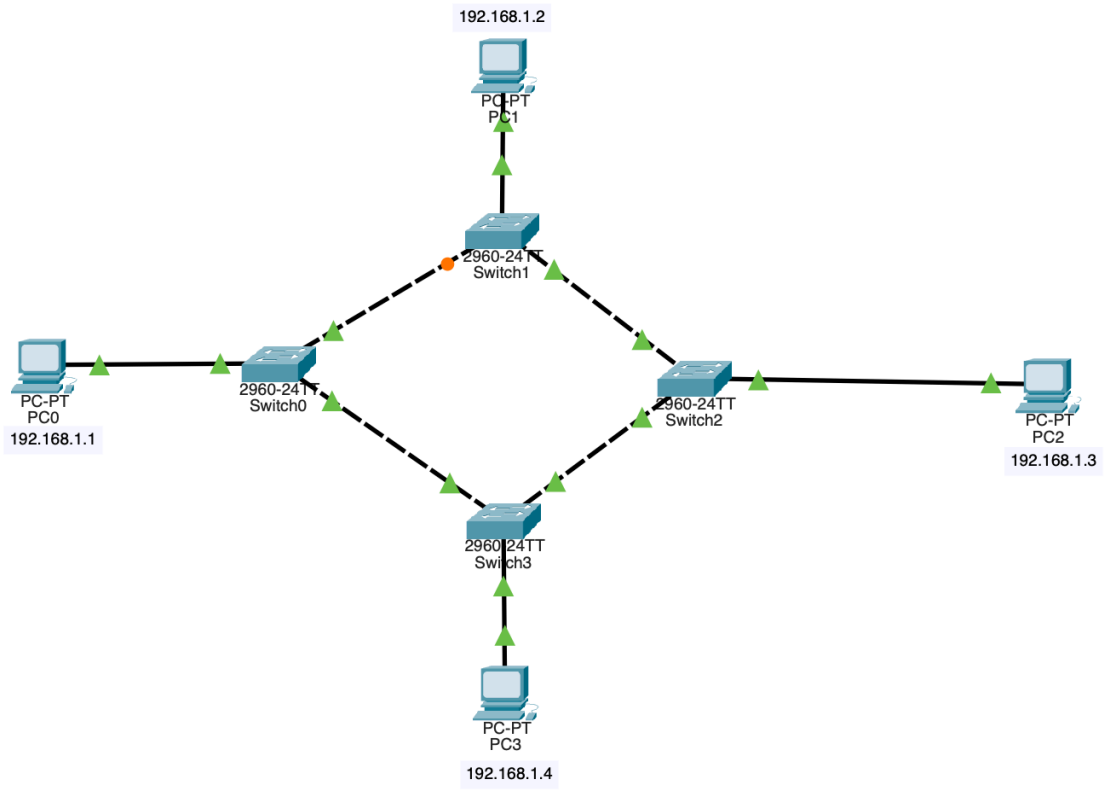
Important distinction: A ring of Ethernet switches in Packet Tracer is not automatically a resilient ring protocol. Switches may use Spanning Tree Protocol (STP) to prevent switching loops by blocking a redundant link.

Practical Implementation in Packet Tracer

Step 1: Create the topology

1. Add four Cisco 2960 switches.
2. Add one PC for each switch.
3. Arrange the switches in a circle or square.
4. Connect each PC to its own switch.

Each PC should connect to its corresponding switch, not directly to another PC.



Observe the link indicators. Because the switches form a physical loop, STP may place one or more ports into a blocking state. This is expected behavior in a typical Layer 2 switched Ethernet network.

Step 2: Configure IP addresses

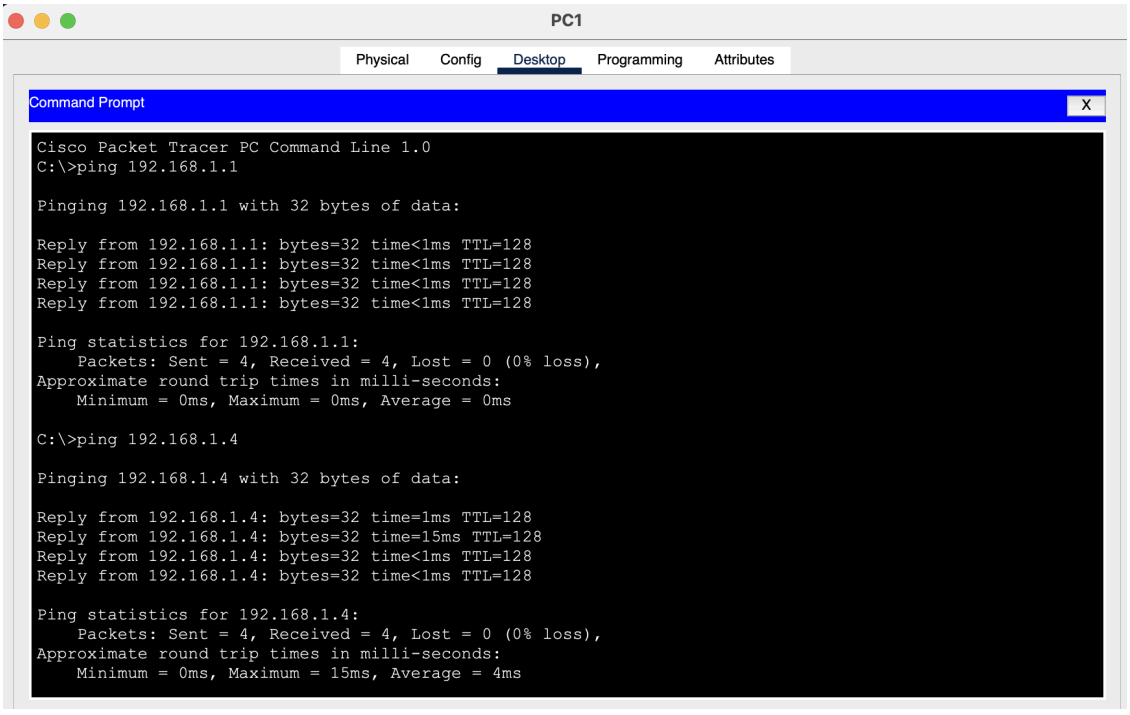
Configure the PCs as follows:

PC	IPv4 Address	Subnet Mask
PC0	192.168.1.1	255.255.255.0
PC1	192.168.1.2	255.255.255.0
PC2	192.168.1.3	255.255.255.0
PC3	192.168.1.4	255.255.255.0

Leave the default gateway blank.

Step 3: Test connectivity

From PC0, run: ping 192.168.1.4. Then test communication with the other PCs.



A switched ring can sometimes maintain communication after one link fails because another path becomes available. However, the outcome depends on STP state and convergence. Do not assume every ring configuration will continue working immediately or without interruption.

Activity 3: Mesh Topology

In a **mesh topology**, devices are connected through multiple links to provide alternative communication paths. For a full mesh of four switches, each switch needs a direct link to the other three switches.

The number of links required for a full mesh is:

$$\textbf{Links} = n(n-1)/2$$

Here, *n* represents the number of devices in the mesh. For four switches:

$$\textbf{Links} = 4 \times 3 / 2 = 6 \textbf{ links}$$

Advantages

- Multiple paths may be available between devices.
- A single link failure may not disconnect the network.
- Can improve resilience when the network is designed appropriately.

Limitations

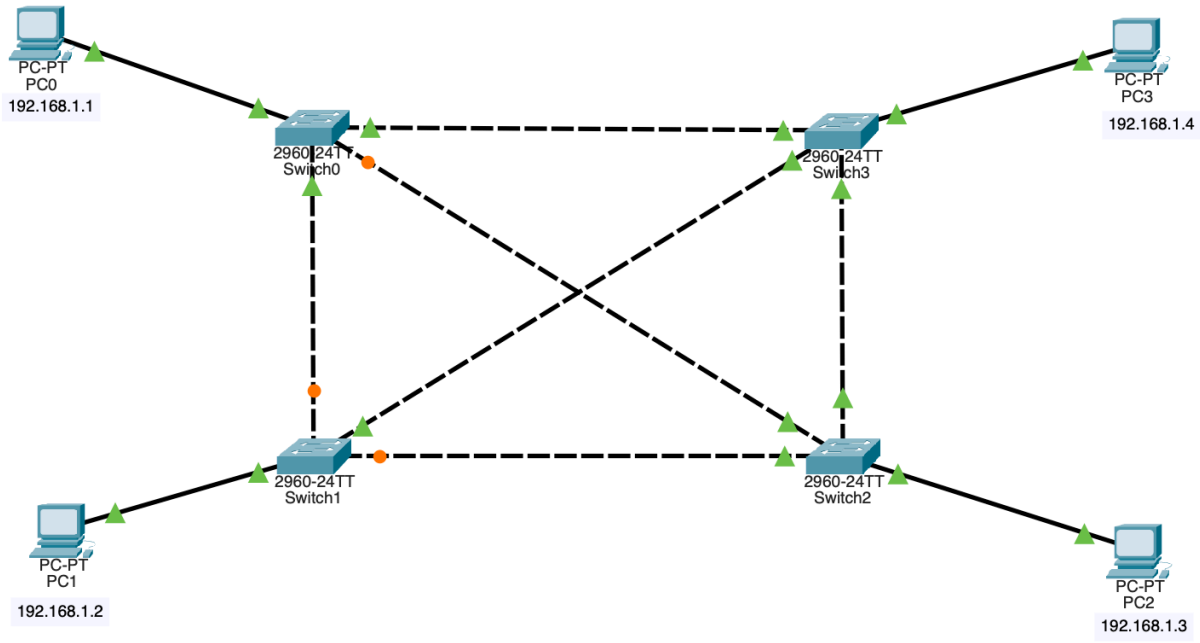
- Requires more cables and switch ports.
- Can be expensive and complex to build.
- Redundant Layer 2 Ethernet links require loop prevention, such as STP.

Practical Implementation in Packet Tracer

Step 1: Create the topology

1. Add four Cisco 2960 switches.
2. Add one PC per switch.
3. Arrange the switches so that the direct connections are easy to see.
4. Connect each PC to its corresponding switch.

5. Each switch should have three direct links to the other switches. Make sure to use different available switch ports for each connection.



Step 2: Configure IP addresses

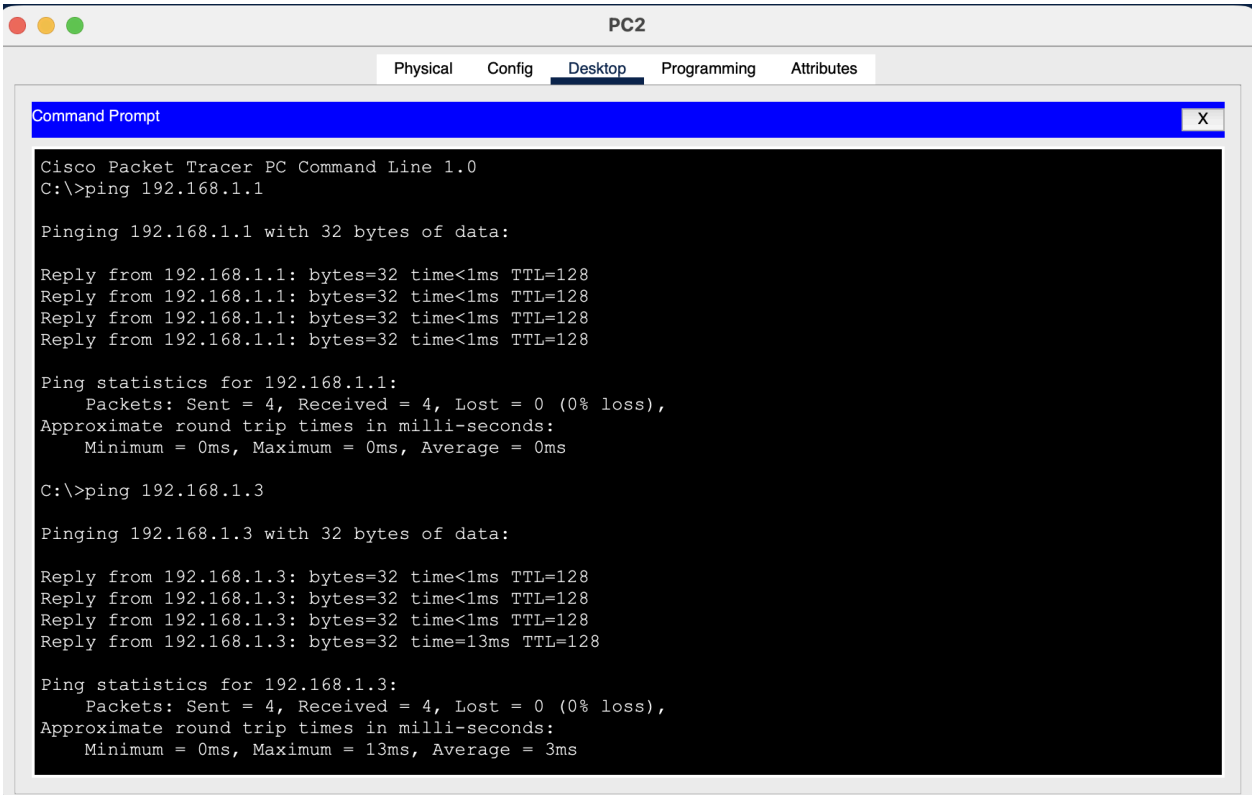
Use the following configuration:

PC	IPv4 Address	Subnet Mask
PC0	192.168.1.1	255.255.255.0
PC1	192.168.1.2	255.255.255.0
PC2	192.168.1.3	255.255.255.0
PC3	192.168.1.4	255.255.255.0

Leave the default gateway blank.

Step 3: Test connectivity

From PC2, execute: ping 192.168.1.4



In a redundant switched Ethernet mesh, STP may block some paths to prevent Layer 2 loops. Consequently, the network has multiple physical links, but not all links necessarily forward ordinary traffic at the same time.

Comparison of the Three Topologies

Star	Ring	Full Mesh
Basic structure		
Devices connect to a central switch	Devices form a closed loop	Every switch connects directly to every other switch
Central device		
Usually present	Not required by the basic ring concept	Not required by the basic mesh concept
Cabling requirement		
Moderate	Depends on the ring size	High

Alternative physical paths		
Limited by central layout	Possible around the other side of the ring	Multiple paths are available
Main weakness		
Central switch failure	A break can disrupt an unprotected ring	Cost and complexity
Packet Tracer observation		
Simple central switch	STP may block a redundant link	STP may block several redundant links